

Skills Summary

Signed Rational Numbers: Mixed Operations

Use this reference while completing your worksheet or when studying.

Skill 1 — Adding signed numbers

Rule: same signs — add the absolute values, keep the sign. Different signs — subtract the absolute values, keep the sign of the number farther from zero.

Example: Find $-8 + 3$

Step 1. The two numbers have different signs, so subtract the absolute values and take the sign of the number farther from zero.

Step 2. $8 - 3 = 5$, and -8 is farther from zero than 3, so the sum is negative.

$$\Rightarrow -8 + 3 = -5$$

Try it: Find $-6 + 10$

check: 4

Skill 2 — Subtracting signed numbers

Rule: $a - b = a + (-b)$ — rewrite every subtraction as adding the opposite, then use the addition rules. Two minus signs in a row make a plus.

Example: Find $-2 - (-9)$

Step 1. The number being subtracted is negative, so change the double negative into addition first, then apply the addition sign rules.

Step 2. $-2 - (-9) = -2 + 9 = 7$, because $9 - 2 = 7$ and 9 is farther from zero.

$$\Rightarrow -2 - (-9) = 7$$

Try it: Find $3 - 8$

check: 5-

Skill 3 — Multiplying and dividing signed numbers

Rule: multiply or divide the absolute values, then set the sign: same signs $\rightarrow$ positive, different signs $\rightarrow$ negative.

Example: Find $(-5)(6)$

Step 1. This is a product, so multiply the absolute values first and decide the sign separately using the same-sign/different-sign rule.

Step 2. $5 \cdot 6 = 30$; the factors have different signs, so the product is negative.

$$\Rightarrow (-5)(6) = -30$$

Try it: Find $-42 \div (-7)$

check: 9

Skill 4 — Mixed operations: choose the method

Rule: work parentheses first, then multiply or divide, then add or subtract. Before comparing or combining, rewrite the numbers in one form (all fractions with a common denominator, or all decimals) — pick whichever form is easier.

Example: Find $-3\left(\frac{1}{2} - 2\right)$

Step 1. Order of operations says to simplify inside the parentheses first, then multiply by the signed factor using the sign rules.

Step 2. $\frac{1}{2} - 2 = -\frac{3}{2}$, and $-3 \cdot \left(-\frac{3}{2}\right)$ has two negative factors, so it is positive: $\frac{9}{2}$.
 $\Rightarrow -3\left(\frac{1}{2} - 2\right) = \frac{9}{2}$

Try it: Find $-2\left(\frac{5}{2} - 3\right)$

check: 1